Saving MQTT Communication Protocol

1、clientid:电表使用自身的表号作为 clientid。

clientid: The electricity meter uses its own meter number as the client ID.

2、Topic

Topic 可设置，默认设置为/v1/device/当前电表表号

(1) 定时数据上报 topic: 当前设置的 topic

(2) 命令下发（电表订阅，平台下发）topic: 当前设置的 topic/cmd

(3) 命令应答 topic: 当前设置的 topic/response

Topic can be configured, default setting is **/v1/device/111111111111（current_meter_id）**

(1) Scheduled data upload Topic: Currently configured Topic

(2) Command dispatch (meter subscription, platform dispatch) Topic: Currently configured Topic/cmd

(3) Command response Topic: Currently configured Topic/response

3、定时数据上报 Scheduled Data Reporting

(1) Topic：当前设置的 topic

Currently set topic

Topic Example：/v1/device/111111111111

(2) 上报字段 Reporting Field

KEY 值	关联变量描述 Associated Variable Description		标签 Label	单位 Unit	数据示例 Sample Data
0	无关联变量	Unrelated variables			
1	A 相电压	A voltage	"Ua"	V	219.9
2	B 相电压	B voltage	"Ub"	V	219.9
3	C 相电压	C voltage	"Uc"	V	219.9
4	A 相电流	A current	“Ia”	A	5.001
5	B 相电流	B current	“Ib”	A	5.001
6	C 相电流	C current	“Ic”	A	5.001
7	总有功功率	Total active power	“Pt”	kW	3.2997
8	A 相有功功率	A active power	“Pa”	kW	1.0999
9	B 相有功功率	B active power	“Pb”	kW	1.0999
10	C 相有功功率	C active power	“Pc”	kW	1.0999
11	总无功功率	Total reactive power	“Qt”	kVar	3.2997
12	A 相无功功率	A reactive power	“Qa”	kVar	1.0999
13	B 相无功功率	B reactive power	“Qb”	kVar	1.0999
14	C 相无功功率	C reactive power	“Qc”	kVar	1.0999
15	总视在功率	Total apparent power	“St”	kVA	3.2997
16	A 相视在功率	A apparent power	“Sa”	kVA	1.0999
17	B 相视在功率	B apparent power	“Sb”	kVA	1.0999
18	C 相视在功率	C apparent power	“Sc”	kVA	1.0999
19	总功率因数	Total Power Factor	“The”		1

20	A 相功率因数	A Power Factor	“TheA”		1
21	B 相功率因数	B Power Factor	“TheB”		1
22	C 相功率因数	C Power Factor	“TheC”		1
23	电网频率	Frequency	“Hz”	Hz	49.99
24	正向有功电量	Forward active energy	kwhImp	kWh	0.01
25	正向有功电量费率 1	Positive Active Energy Rate 1	kwhImpTariff1	kWh	0.01
26	正向有功电量费率 2	Positive Active Energy Rate 2	kwhImpTariff2	kWh	0
27	正向有功电量费率 3	Positive Active Energy Rate 3	kwhImpTariff3	kWh	0
28	正向有功电量费率 4	Positive Active Energy Rate 4	kwhImpTariff4	kWh	0
29	反向有功电量	Reverse active energy	kwhExp	kWh	0.01
30	反向有功电量费率 1	Reverse Active Energy Rate 1	kWhExpTariff1	kWh	0.01
31	反向有功电量费率 2	Reverse Active Energy Rate 2	kwhExpTariff2	kWh	0
32	反向有功电量费率 3	Reverse Active Energy Rate 3	kwhExpTariff3	kWh	0
33	反向有功电量费率 4	Reverse Active Energy Rate 4	kwhExpTariff4	kWh	0
34	时间日期	Time and Date	Time		2021-08-20 20:30:16
35	电表表号	Meter Address	Device_Number		111111111111
36	报文类型	Message Type	DataType		
37	峰值电流	Peak current	PeakIa	A	上 1 分钟的最大电流值，每 1 分钟更新一次数值
38	连续电流	Continuous current	ContinuousIa	A	上 5 分钟的电流平均值，每 5 分钟更新一次数值
39	峰值电压	Peak voltage	PeakUa	V	上 1 分钟的最大电压值，每 1 分钟更新一次数值

(3) 定时数据上报示例：Example of Scheduled Data Reporting

{"DataType":"Variable","Device_Number":"111111111111","Time":"2021-08-20 20:30:16","datas":{"Ua":647.4,"Ub":0,"Uc":0,"Ia":0,"Ib":0,"Ic":0,"Pt":0,"Pa":0,"Pb":0,"Pc":0,"The":0.02,"TheA":0.02,"TheB":1,"TheC":1,"Hz":49.99,"kWhImp":0,"kWhImpTariff1":0,"kWhImpTariff2":0,"kW hImpTariff3":0,"kWhImpTariff4":0,"kWhExp":0,"kWhExpTariff1":0,"kWhExpTariff2":0,"kWhExpTariff3":0,"kWhExpTariff4":0}}

4、命令下发(电表订阅，平台下发)

Command Issuance (Electric Meter Subscription, Platform Issuance)

(1)topic：Current settings topic/cmd

Topic Example：/v1/device/111111111111/cmd

(2) 命令下发字段 Command Issuance Field

KEY 值	关联变量描述 Associated Variable Description	标签	单位	数据示例
0	无关联变量 Unrelated variables			
1	电表表号 Meter Address	"meteraddress"		111111111111
2	命令 ID Command ID	"messageID"		
3	命令 Command	"cmd"		“relaycontrol” “getrelaystatus” “readvariable” “setip” “getip” “setfreq” “getfreq” “setmqttpara” “getmqttpara”
4	执行结果 Execution Results	“cmdresult”		0:成功 1:失败
5	继电器状态 Relay Status	“relaystatus”		“on”:合闸 “off”:拉闸

(3) 获取继电器状态 Retrieve relay status：

下发命令 Issue an order：

{"meteraddress":"111111111111","messageID":"2022121013091201","cmd":"getrelaystatus"}

应答 Response：

{"meteraddress":"111111111111","messageID":"2022121013091201","cmdresult":0,”cmd”:”getrelaystatus”,”relaystatus”:”on”}

(4) 拉合闸命令 Close the circuit breaker

message 内容为 relayon 时，电表合闸。message 内容为 relayoff 时，电表拉闸

When the message content is **relayon**, the meter energizes.

When the message content is **relayoff,** the meter de-energizes.

拉闸命令示例 Example of a power cut-off command：

{"meteraddress":"230920020021","messageID":"1234560",”cmd”:”relaycontrol”,"message":"relayoff"}

合闸命令示例 Closing Command Example：

{"meteraddress":"230920020021","messageID":"1234560",”cmd”:”relaycontrol”,"message":"relayon"}

(5) 抄读实时数据 Read real-time data

命令 Command：

{"meteraddress":"111111111111","messageID":"2022121013091201","cmd":"readvariable"}

应答 Response：

{"meteraddress":"111111111111","messageID":"2022121013091201","cmdresult":0,”cmd”:”readvariable”,”datas”:{"Ua":647.4,"Ub":0,"Uc":0,"Ia":0,"Ib":0,"Ic":0,"Pt":0,"Pa":0,"Pb":0,"Pc":0,"The":0.02,"TheA":0.02,"TheB":1,"TheC":1,"Hz":49.99,"kWhImp":0,"kWhImpTariff1":0,"kWhImpTariff2":0,"kWhImpTariff3":0,"kWhImpTariff4":0,"kWhExp":0,"kWhExpTariff1":0,"kWhExpTariff2":0,"kWhExpTariff3":0,"kWhExpTariff4":0}}

(6) 设置 IP 端口 Set IP Port

命令 Command：

{"meteraddress":"165125062053","messageID":"2022121013091201","cmd":"setip","HOST":"117.84.15.151","PORT":18888}

应答 Response：

meteraddress":"165125062053","messageID":"2022121013091201","cmdresult":"0","cmd":"setip"}

(7) 抄读 IP 端口 Read IP port

命令 Command：

{"meteraddress":"165125062053","messageID":"2022121013091201","cmd":"getip"}

应答 Response：

{"meteraddress":"165125062053","messageID":"2022121013091201","cmdresult":"0","cmd":"getip","HOST":"39.104.65.88","PORT":1883}

(8) 设置上报周期 Set the reporting cycle

命令 Command：

{"meteraddress":"165125062053","messageID":"2022121013091201","cmd":"setfreq","FREQ":900}

应答 Response：

{"meteraddress":"165125062053","messageID":"2022121013091201","cmdresult":"0","cmd":"setfreq"}

(9) 抄读上报周期 Reading Period

命令 Command：

{"meteraddress":"111111111111","messageID":"2022121013091201","cmd":"getfreq"}

应答 Response：

{"meteraddress":"111111111111","messageID":"2022121013091201","cmdresult":0,”cmd”:”getfreq”,”FREQ”:300}

(10) 设置 MQTT 连接用户名、密码，上报 topic
Set the MQTT connection username and password, and configure the Topic.

命令 Command：

{"meteraddress":"111111111111","messageID":"2022121013091201","cmd":"setmqttpara","USERNAME":"SAV","PASSWORD":"123456","TOPIC":"v1/devide/123456"}

应答主题使用原 topic/response
Use the original topic/response for replies.

应答 Response：

{"meteraddress":"111111111111","messageID":"2022121013091201","cmdresult":0,"cmd":"setmqttpara"}

(11) 抄读 MQTT 连接用户名、密码 Reading MQTT Connection Username and Password

命令 Command：

{"meteraddress":"111111111111","messageID":"2022121013091201","cmd":"getmqttpara"}

应答 Response：

{"meteraddress":"165125062053","messageID":"2022121013091201","cmdresult":"0","cmd":"getmqttpara","USERNAME":"sav","PASSWORD":"XUMQTT"}

(12) 设置时间日期 Set Time and Date

命令 Command：

{"meteraddress":"165125062053","messageID":"2022121013091201","cmd":"setdatetime","DATETIME":"2025-09-02T15:00:00"}

应答 Response：

{"meteraddress":"165125062053","messageID":"2022121013091201","cmdresult":0,"cmd":"set datetime"}

(13) 抄读时间日期 Read time and date

命令 Command：

{"meteraddress":"165125062053","messageID":"2022121013091201","cmd":"getdatetime"}

应答 Response：

{"meteraddress":"165125062053","messageID":"2022121013091201","cmdresult":"0","cmd":"getdatetime","DATETIME":"2025-09-02T15:13:00"}

(14) 抄读历史记录 Copy Reading History

命令 Command：

{"meteraddress":"210924010003","messageID":"2022121013091201","cmd":"gethistorydata","BEGINTIME":"2025-10-13T15:00:00","ENDTIME":"2025-10-14T14:40:00"}

应答 Response：

subsequentFalg：代表是否有后续帧，0：数据帧结束 1：还有后续帧

frameNO:帧序号

（subsequentFalg: Indicates whether subsequent frames exist. 0: Data frame end. 1: Subsequent frames follow.

frameNO: Frame sequence number）

{"messageID":"2022121013091201","meteraddress":"210924010003","cmdresult":"0","cmd":"gethistorydata","frameNO":1,"historydata":[{"20251013150000":0},{"20251013151000":0.16},{"20251013155000":0.4},{"20251013160000":0.66},{"20251013161000":0.92},{"20251013162000":1.18},{"20251013163000":1.44},{"20251013164000":1.7},{"20251013165000":1.96},{"20251013170000":2.22},{"20251013171000":2.48},{"20251013172000":2.74},{"20251013173000":3.01},{"20251013174000":3.27},{"20251013175000":3.53}],"subsequentFalg":1}

{"messageID":"2022121013091201","meteraddress":"210924010003","cmdresult":"0","cmd":"gethistorydata","frameNO":2,"historydata":[{"20251013180000":3.8},{"20251013181000":4.06},{"20251013182000":4.33},{"20251013183000":4.59},{"20251013184000":4.86},{"20251013185000":5.12},{"20251013190000":5.38},{"20251013191000":5.64},{"20251013192000":5.91},{"20251013193000":6.17},{"20251013194000":6.43},{"20251013195000":6.7},{"20251013200000":6.96},{"20251013201000":7.22}],"subsequentFalg":1}

{"messageID":"2022121013091201","meteraddress":"210924010003","cmdresult":"0","cmd":"gethistorydata","frameNO":3,"historydata":[{"20251013202000":7.49},{"20251013203000":7.75},{"20251013204000":8.02},{"20251013205000":8.28},{"20251013210000":8.55},{"20251013211000":8.81},{"20251013212000":9.08},{"20251013213000":9.35},{"20251013214000":9.61},{"20251013215000":9.88},{"20251013220000":10.14},{"20251013221000":10.4},{"20251013222000":10.66},{"20251013223000":10.92}],"subsequentFalg":1}

{"messageID":"2022121013091201","meteraddress":"210924010003","cmdresult":"0","cmd":"gethistorydata","frameNO":4,"historydata":[{"20251013224000":11.18},{"20251013225000":11.44},{"2025

1013230000":11.7},{"20251013231000":11.96},{"20251013232000":12.23},{"20251013233000":12.4
9},{"20251013234000":12.75},{"20251013235000":13.02},{"20251014000000":13.28},{"2025101400
1000":13.55},{"20251014002000":13.81},{"20251014003000":14.07},{"20251014004000":14.33},{"2
0251014005000":14.59}],"subsequentFalg":1}

{"messageID":"2022121013091201","meteraddress":"210924010003","cmdresult":"0","cmd":"gethis
torydata","frameNO":5,"historydata":[{"20251014010000":14.86},{"20251014011000":15.12},{"2025
1014012000":15.38},{"20251014013000":15.64},{"20251014014000":15.9},{"20251014015000":16.1
7},{"20251014020000":16.43},{"20251014021000":16.69},{"20251014022000":16.95},{"2025101402
3000":17.21},{"20251014024000":17.47},{"20251014025000":17.74},{"20251014030000":18},{"2025
1014031000":18.27}],"subsequentFalg":1}

{"messageID":"2022121013091201","meteraddress":"210924010003","cmdresult":"0","cmd":"gethis
torydata","frameNO":6,"historydata":[{"20251014032000":18.53},{"20251014033000":18.8},{"20251
014034000":19.06},{"20251014035000":19.33},{"20251014040000":19.59},{"20251014041000":19.8
6},{"20251014042000":20.12},{"20251014043000":20.38},{"20251014044000":20.64},{"2025101404
5000":20.9},{"20251014050000":21.17},{"20251014051000":21.43},{"20251014052000":21.7},{"202
51014053000":21.96}],"subsequentFalg":1}

{"messageID":"2022121013091201","meteraddress":"210924010003","cmdresult":"0","cmd":"gethis
torydata","frameNO":7,"historydata":[{"20251014054000":22.23},{"20251014055000":22.49},{"2025
1014060000":22.76},{"20251014061000":23.02},{"20251014062000":23.28},{"20251014063000":23.
54},{"20251014064000":23.8},{"20251014065000":24.07},{"20251014070000":24.33},{"2025101407
1000":24.59},{"20251014072000":24.85},{"20251014073000":25.11},{"20251014074000":25.38},{"2
0251014075000":25.65}],"subsequentFalg":1}

{"messageID":"2022121013091201","meteraddress":"210924010003","cmdresult":"0","cmd":"gethis
torydata","frameNO":8,"historydata":[{"20251014080000":25.91},{"20251014081000":26.18},{"2025
1014082000":26.45},{"20251014083000":26.71},{"20251014084000":26.97},{"20251014085000":27.
24},{"20251014090000":27.51},{"20251014091000":27.77},{"20251014092000":28.04},{"202510140
93000":28.3},{"20251014094000":28.56},{"20251014095000":28.82},{"20251014100000":29.08},{"2
0251014101000":29.34}],"subsequentFalg":1}

{"messageID":"2022121013091201","meteraddress":"210924010003","cmdresult":"0","cmd":"gethis
torydata","frameNO":9,"historydata":[{"20251014102000":29.6},{"20251014103000":29.83},{"20251
014104000":30},{"20251014105000":30.3},{"20251014110000":30.6},{"20251014111000":30.9},{"20
251014112000":31.2}],"subsequentFalg":0}